

Andres Perez

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EDUCATION

University of Notre Dame | Notre Dame, IN

Graduation Date: May 2028

Bachelor of Science: Computer Engineering | Minor: Engineering Corporate Practice

GPA: 3.7

Engineering Study Abroad: London, England

May 2025 – June 2025

Relevant Courses: Data Structures, Discrete Math, Intro. to Electrical and Computer Engineering, Intro. To Embedded Systems, Linear Algebra & Differential Equations, Engineering Physics II, Integrated Engineering and Business Fundamentals

ENGINEERING LEADERSHIP

IrishSat's Gravitational Orbital Attitude Thermal Lab (GOAT Lab)

Notre Dame, IN

Electronics Lead | CubeSat Team

Aug. 2024 – Present

- Spearheading a team of 6+ engineers for the embedded systems development and PCB design initiatives for testing equipment, particularly a Helmholtz cage project, coordinating hardware-software integration across CubeSat projects.
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, integrating PCB design (KiCad), control algorithms (MATLAB), and embedded Linux systems (Raspberry Pi) into a unified testing platform.
- Assembled a custom 2S2P battery pack using 18650 lithium-ion cells to make a 7.2V 7000 mAh battery pack. The completed pack was then evaluated with a Battery Management System to validate its nominal values.

ENGINEERING PROJECTS

Hardware-Accelerated Human Activity Classifier | Independent Project

Long Beach, CA

Designer

July 2025 – Aug. 2025

- Built a wireless data acquisition system using two Arduino Uno R3s, an MPU-9250 IMU sensor, and nRF24L01 RF transceivers.
- Developed a C++ sketch to sample accelerometer and gyroscope data and transmitted it wirelessly to a receiving station. Along with a Python script to receive, parse the real-time sensor data, generating a custom labeled dataset for walking and sitting.
- Trained a 1D Convolutional Neural Network in PyTorch to classify human activities with 96.7% accuracy, documenting model stability and performance by visualizing accuracy against hyperparameters (learning rate, window size, and optimizer).

Helmholtz Cage Driver PCB | GOAT Lab

Notre Dame, IN

Lead Designer

Mar. 2025

- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino, magnetometers, and H-bridge circuits, reducing assembly complexity.
- Amended the generated Bill of Materials and Component Placement List files export to JLCPCB for manufacturing.

Air Quality Sensor Housing | Engineering Design

Notre Dame, IN

Lead Designer

Dec. 2024

- Collaborated with a team of four engineering students to develop and iterate on sensor housing with a rack-and-pinion opening mechanism actuated by a servo motor and ESP32-based microcontroller.
- Designed and prototyped with SolidWorks in two design phases, optimizing for form, fit, function, and durability through iterative testing and feedback.

Portfolio Website | Independent Project | andres-perez.github.io

Long Beach, CA

Developer

July 2024– Present

- Developed a student and engineering project portfolio website with Astro and Tailwind CSS frameworks as the front end.

RESEARCH & WORK EXPERIENCE

Student Researcher | Notre Dame Department of Electrical Engineering

Aug. 2025 – Present

- Developing a real-time Python software interface for tracking and visualizing GPS-enabled test resources, for drones and software-defined radios, that parses NMEA strings to extract GPS data and displays it on a live map using the folium library, supporting various research experiments.

Engineering Career Assistant | Meruelo Family Center for Career Development

Sept. 2025 – Present

- Coauthoring a bi-weekly newsletter for sophomores, juniors, and seniors, highlighting career opportunities and alumni spotlights from various majors and post-graduate experiences.
- Grew the letter to a best 76.3% open rate with a 2.3% click rate of 517 recipients.

TECHNICAL SKILLS, LANGUAGE & INTERESTS

Hardware & Tools: KiCad, Wireshark, SolidWorks, Oscilloscope, Soldering, PCB Assembly

Programming: C/C++, Python (Threading, Socket, PyTorch, Pandas, NumPy), Git, GitHub, MATLAB, Astro

Interests: Embedded Systems, Machine Learning, Data Acquisition, Avionics, PCB Design

Skills: Microsoft Office Suite (Excel, PowerPoint, Word), Vim, Technical Writing

Language: English (Native), Spanish (Native), Japanese (Elementary)